

Shubham Singh

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Summary

Full-stack software engineer at Amazon with ~4 years shipping **React/TypeScript** frontends, **Python/Go** backend services, and **AWS** production systems. Led cross-team design of an internal developer platform powering **2,000+ compliance launches**; built APIs and workflows for hybrid-retrieval Q&A, distributed orchestration, and large-scale data processing.

Experience

Amazon

Software Development Engineer II

Software Development Engineer I

Aug 2022 – Present

Aug 2023 – Present

Aug 2022 – Aug 2023

- **Led cross-team design** of an internal compliance control-launch **developer platform (React, TypeScript, Redux + AWS API Gateway/Lambda/DynamoDB)**; **sole author of the system HLD**, coordinating **3 partner teams across 3 timezones**; powers **2,000+ compliance control launches** and reduces multi-week onboarding to self-serve hours.
- Built **three net-new PO UX surfaces** on the platform with **React, Redux, and RTK Query**; integrated the v2 Control Plane API with a type-safe client and implemented **role-gated rendering across 5 user roles**, covered by component tests.
- Built and shipped **MaverickIQ**, an agentic Q&A service over **hybrid BM25 + kNN retrieval** on AWS Bedrock with secure tool-calling, citation-grounded outputs, and audit logging; deflected **~90%** of manual interpretation requests across **5 partner teams**.
- Designed a **stateful workflow orchestration platform** (Step Functions + DynamoDB) replacing brittle Lambda chains: per-shard Map parallelism, callbacks for long-running operations, lock-based resource ownership, and bounded compensation; cut workflow failure rate from **8%** to **<1%**.
- Designed a **PySpark data pipeline** processing **30M+ monthly records** with schema versioning, validation, and error isolation; reduced incident resolution time by **70%**.
- Owned production operations across **5 services** on rotating on-call, drove **root-cause analyses** across high-severity incidents, and mentored a new-grad SDE to independent on-call readiness through incident-triage pairing and design-review coaching.

Open Source & Projects

Agent Replay (*React 19, TypeScript, Canvas, Web Workers, Zustand*): browser-based time-travel debugger for LLM agent runs. Scrub a trace like a video on a Canvas waterfall, replay token streams, and diff two runs. Renders **50k+ spans at 60fps** via structure-of-arrays geometry, viewport culling, and LOD aggregation. **2026**

AgentOps RAG (*Python, FastAPI, OpenSearch, Redis*): production-style RAG service with FastAPI routes, JWT auth, async workers, hybrid retrieval, and OTel observability. **2026**

PostHog/code OSS Contribution (Merged) (*TypeScript, Electron*): shipped terminal rendering fix into PostHog's coding-agent app: UTF-8 PTY decoding, xterm font alignment, and parameterized tests across both spawn paths. **2026**

Technical Skills

Frontend: React, TypeScript, Redux, RTK Query, Vite, MDX, CSS Modules, responsive layouts, accessibility (ARIA), Lighthouse perf

Backend: Python (FastAPI), Go, Java, Node.js, REST APIs, Microservices, Event-Driven Architecture

AWS/Infra: Lambda, Step Functions, DynamoDB, S3, API Gateway, CloudFront, Docker, Kubernetes, CI/CD

Data/AI: PySpark, Postgres, Redis, OpenSearch, RAG, Hybrid Retrieval (BM25+kNN), AWS Bedrock, Tool-Calling

Tooling: Git, GitHub Actions, OpenTelemetry, Playwright, Jest, ESLint

Education

Indiana University Bloomington

M.S. in Data Science

Jan 2021 – Jul 2022

Bloomington, IN